



CANDIDATE
NAME

--

CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--

9709/52

June 2026

1 hour 15 minutes

You will need: List of formulae (MF19)

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **14** pages. Any blank pages are indicated.

- 1 Jacob throws three coins at the same time.
The first coin is biased so that the probability of obtaining a head when it is thrown is $\frac{1}{3}$.
The second coin is biased so that the probability of obtaining a head when it is thrown is $\frac{1}{4}$.
The third coin is biased so that the probability of obtaining a head when it is thrown is $\frac{1}{5}$.
The random variable X is the number of heads obtained.

(a) Show that $P(X = 2) = \frac{3}{20}$. [1]

.....

.....

.....

.....

.....

.....

.....

.....

.....

(b) Draw up the probability distribution table for X . [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[2]

[illegible]

- 2 Last year, an online store sold a large number of computers. 55% of the computers were made by company F , 30% were made by company G and 15% were made by company H .

A random sample of 3 customers who each bought a computer from this store is chosen.

- (a) Find the probability that the 3 customers bought computers all made by different companies. [1]

.....

.....

.....

.....

.....

.....

.....

.....

A random sample of 12 customers who each bought a computer from this store is chosen.

- (b) Find the probability that fewer than 10 of these customers bought a computer made by company F . [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(c) Use a suitable approximation to find the probability that more than 24 of these customers bought a computer made by company H . [5]

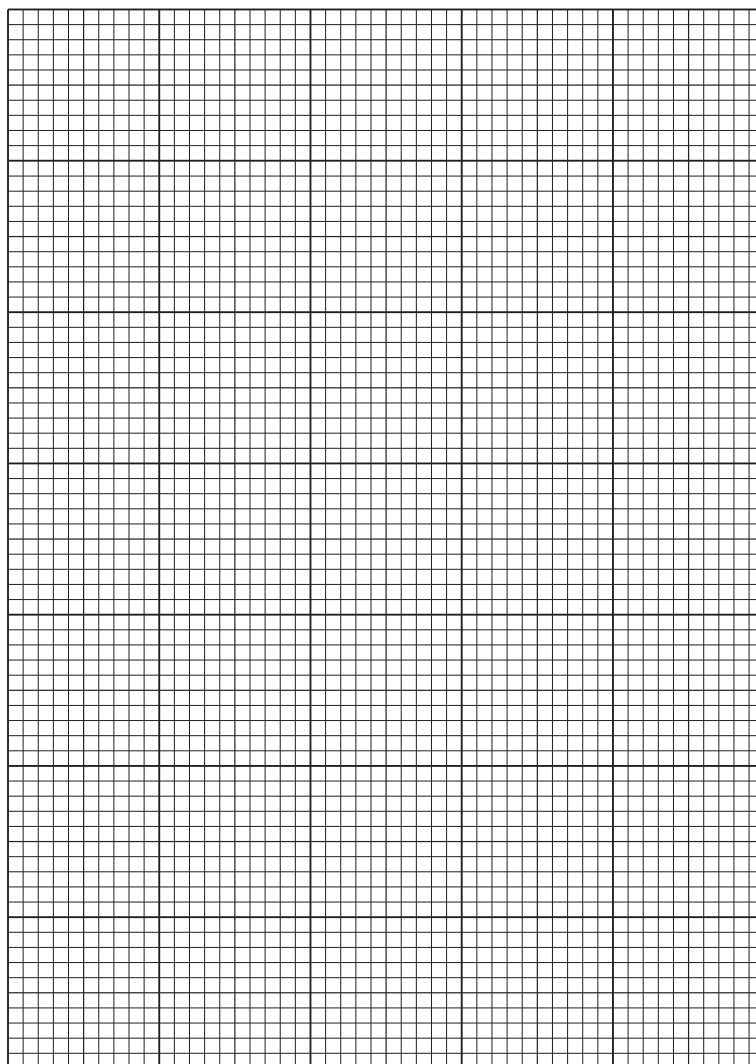
[illegible]

- 3 The lengths of 250 leaves of a certain type of plant are measured, correct to the nearest centimetre. The results are summarised in the table below.

Length (cm)	5 – 9	10 – 14	15 – 19	20 – 24	25 – 29	30 – 39
Frequency	18	28	60	72	48	24

- (a) On the grid below, draw a cumulative frequency graph to illustrate this information.

[4]



- [2]

.....

.....

.....

.....

.....

- [3]

[illegible]

(a) Find the probability that all three cars are the same colour. [3]

[illegible]

[4]

[illegible]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

The mass of cherries sold per day in a supermarket is normally distributed with mean 72.4 kg and standard deviation σ kg. It is known that on 10% of days less than 59.1 kg of cherries are sold.

- (b) Find the value of σ . [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

The supermarket is open 7 days a week.

- (c) Find the probability that, in a randomly chosen week, the first day on which less than 59.1 kg of cherries are sold is the fifth day of the week. [1]

.....

.....

.....

- (d) Find the probability that, in a randomly chosen week, the first day on which less than 59.1 kg of cherries are sold is before the fifth day of the week. [2]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- Alissa places the books in a row on her shelf. She is only interested in the arrangement of the colours.

[illegible]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

- (c) How many different colour arrangements are there of the 10 books with exactly 4 books between the 2 yellow books? [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Alissa selects 4 books from her 10 different books from the series Squares and Circles.

- (d) Find the number of different selections if the 4 books include at least 1 red book, at most 1 blue book and exactly 1 yellow book. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

14

